

# Trig identities

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Definitions:

$$\begin{aligned}\tan(x) &= \sin(x)/\cos(x) \\ \csc(x) &= 1/\sin(x) \\ \sec(x) &= 1/\cos(x) \\ \cot(x) &= 1/\tan(x)\end{aligned}$$

Pythagorean:

$$\begin{aligned}\sin^2(x) + \cos^2(x) &= 1 \\ 1 + \cot^2(x) &= \csc^2(x) \\ 1 + \tan^2(x) &= \sec^2(x)\end{aligned}$$

Cofunction:

$$\begin{aligned}\sin(\pi/2 - x) &= \cos(x) & \cos(\pi/2 - x) &= \sin(x) \\ \tan(\pi/2 - x) &= \cot(x) & \cot(\pi/2 - x) &= \tan(x) \\ \sec(\pi/2 - x) &= \csc(x) & \csc(\pi/2 - x) &= \sec(x)\end{aligned}$$

Difference/sum:

$$\begin{aligned}\cos(\alpha - \beta) &= \cos(\alpha)\cos(\beta) + \sin(\alpha)\sin(\beta) \\ \cos(\alpha + \beta) &= \cos(\alpha)\cos(\beta) - \sin(\alpha)\sin(\beta) \\ \sin(\alpha - \beta) &= \sin(\alpha)\cos(\beta) - \cos(\alpha)\sin(\beta) \\ \sin(\alpha + \beta) &= \sin(\alpha)\cos(\beta) + \cos(\alpha)\sin(\beta) \\ \tan(\alpha - \beta) &= \frac{\tan(\alpha) - \tan(\beta)}{1 + \tan(\alpha)\tan(\beta)} \\ \tan(\alpha + \beta) &= \frac{\tan(\alpha) + \tan(\beta)}{1 - \tan(\alpha)\tan(\beta)}\end{aligned}$$

Double angle:

$$\begin{aligned}\sin(2x) &= 2\sin(x)\cos(x) \\ \tan(2x) &= \frac{2\tan(x)}{1 - \tan^2(x)} \\ \cos(2x) &= \cos^2(x) - \sin^2(x) \\ &= 2\cos^2(x) - 1 \\ &= 1 - 2\sin^2(x)\end{aligned}$$

Half angle:

$$\begin{aligned}\sin(x/2) &= \pm\sqrt{1/2(1 - \cos(x))} \\ \cos(x/2) &= \pm\sqrt{1/2(1 + \cos(x))} \\ \tan(x/2) &= \frac{1 - \cos(x)}{\sin(x)} \\ &= \frac{\sin(x)}{1 + \cos(x)}\end{aligned}$$

Product/sum:

$$\begin{aligned}2\cos(\alpha)\cos(\beta) &= \cos(\alpha - \beta) + \cos(\alpha + \beta) \\ 2\sin(\alpha)\sin(\beta) &= \cos(\alpha - \beta) - \cos(\alpha + \beta) \\ 2\sin(\alpha)\cos(\beta) &= \sin(\alpha + \beta) + \sin(\alpha - \beta) \\ 2\cos(\alpha)\sin(\beta) &= \sin(\alpha + \beta) - \sin(\alpha - \beta)\end{aligned}$$

Sum/product:

$$\begin{aligned}\sin(2\alpha) + \sin(2\beta) &= 2\sin(\alpha + \beta)\cos(\alpha - \beta) \\ \sin(2\alpha) - \sin(2\beta) &= 2\cos(\alpha + \beta)\sin(\alpha - \beta) \\ \cos(2\alpha) + \cos(2\beta) &= 2\cos(\alpha + \beta)\cos(\alpha - \beta) \\ \cos(2\alpha) - \cos(2\beta) &= -2\sin(\alpha + \beta)\sin(\alpha - \beta)\end{aligned}$$

Power reducing:

$$\begin{aligned}\sin^2(x) &= (1 - \cos(2x))/2 \\ \cos^2(x) &= (1 + \cos(2x))/2 \\ \tan^2(x) &= \frac{1 - \cos(2x)}{1 + \cos(2x)}\end{aligned}$$

Hyperbolic:

$$\begin{aligned}\sinh(iz) &= i\sin(z), & \sin(iz) &= i\sinh(z) & (z \in \mathbb{R}) \\ \cosh(iz) &= \cos(z), & \cos(iz) &= \cosh(z) & (z \in \mathbb{R}) \\ \cosh^2(z) - \sinh^2(z) &= 1 & (z \in \mathbb{C})\end{aligned}$$

Exponential formulae:

$$\begin{aligned}\sin(z) &= (e^{iz} - e^{-iz})/(2i) \\ \cos(z) &= (e^{iz} + e^{-iz})/2 \\ \sinh(z) &= (e^z - e^{-z})/2 \\ \cosh(z) &= (e^z + e^{-z})/2\end{aligned}$$

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\*last updated Jan 30 2014, and Oct 8 2026 with plots

## Derivatives:

$$\begin{aligned}
 \frac{d}{dx} \sin(x) &= \cos(x) dx \\
 \frac{d}{dx} \cos(x) &= -\sin(x) dx \\
 \frac{d}{dx} \tan(x) &= \sec^2(x) dx \\
 \frac{d}{dx} \cot(x) &= -\csc^2(x) dx \\
 \frac{d}{dx} \sec(x) &= \sec(x) \tan(x) dx \\
 \frac{d}{dx} \csc(x) &= -\csc(x) \cot(x) dx \\
 \frac{d}{dx} \sin^2(x) &= 2 \sin(x) \cos(x) dx \\
 \frac{d}{dx} \cos^2(x) &= -2 \sin(x) \cos(x) dx \\
 \frac{d}{dx} \tan^2(x) &= 2 \tan(x) \sec^2(x) dx \\
 \frac{d}{dx} \arccos(x) &= -(1-x^2)^{-\frac{1}{2}} dx \\
 \frac{d}{dx} \arcsin(x) &= (1-x^2)^{-\frac{1}{2}} dx \\
 \frac{d}{dx} \arctan(x) &= (1+x^2)^{-1} dx \\
 \frac{d}{dx} \cosh(x) &= \sinh(x) dx \\
 \frac{d}{dx} \sinh(x) &= \cosh(x) dx
 \end{aligned}$$

## Maclaurin Series (and Laurent Series about origin):

$$\begin{aligned}
 e^x &= \sum_{n=0}^{\infty} \frac{x^n}{n!} \\
 \sin(x) &= \sum_{n=0}^{\infty} \frac{(-1)^{n-1}}{(2n-1)!} x^{2n-1} \\
 &= x - x^3/3! + x^5/5! - \dots \\
 \cos(x) &= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} \\
 &= 1 - x^2/2! + x^4/4! - \dots \\
 \tan(x) &= x + 1/2 \cdot x^3 + 2/15 \cdot x^5 + 17/315 \cdot x^7 + \dots \\
 &= \sum_{n=1}^{\infty} \frac{B_{2n}(-4)^n(1-4^n)}{(2n)!} x^{2n-1} \quad \text{for } |x| < \frac{\pi}{2} \\
 &[B_n \text{ is the } n^{th} \text{ Bernoulli number}] \\
 \cot(z) &= z^{-1} - z/3 - z^3/45 - 2/945 \cdot z^5 - \dots \\
 &[\text{for } 0 < |z| < \pi] \\
 \csc(z) &= z^{-1} + z/6 + 7/360 \cdot z^3 + .002 \cdot z^5 + \dots \text{ for } 0 < |z| < \pi \\
 \sec(z) &= 1 + z^2/2 + 5/24 \cdot z^4 + \dots \text{ for } |z| < \pi/2
 \end{aligned}$$

## Misc. Functions

### Sine Cardinal:

$$\text{sinc}(x) = \begin{cases} 1 & \text{if } x = 0 \\ \frac{\sin(x)}{x} & \text{otherwise} \end{cases}$$

### Sine Integral:

$$\text{Si}(z) = -\int_0^z \frac{\sin(t)}{t} dt$$

## Integrals

$$\begin{aligned}
 \int_0^x \cos^2(\theta) d\theta &= 1/2x + 1/4 \sin(2x) \\
 \int_0^x \sin^2(\theta) d\theta &= 1/2x - 1/4 \sin(2x) \\
 \int_0^\pi \sin^2(x) dx &= \pi/2 \\
 &= \int_0^\pi \cos^2(x) dx \\
 \int_0^{2\pi} \sin^2(x) dx &= \pi \\
 &= \int_0^{2\pi} \cos^2(x) dx
 \end{aligned}$$

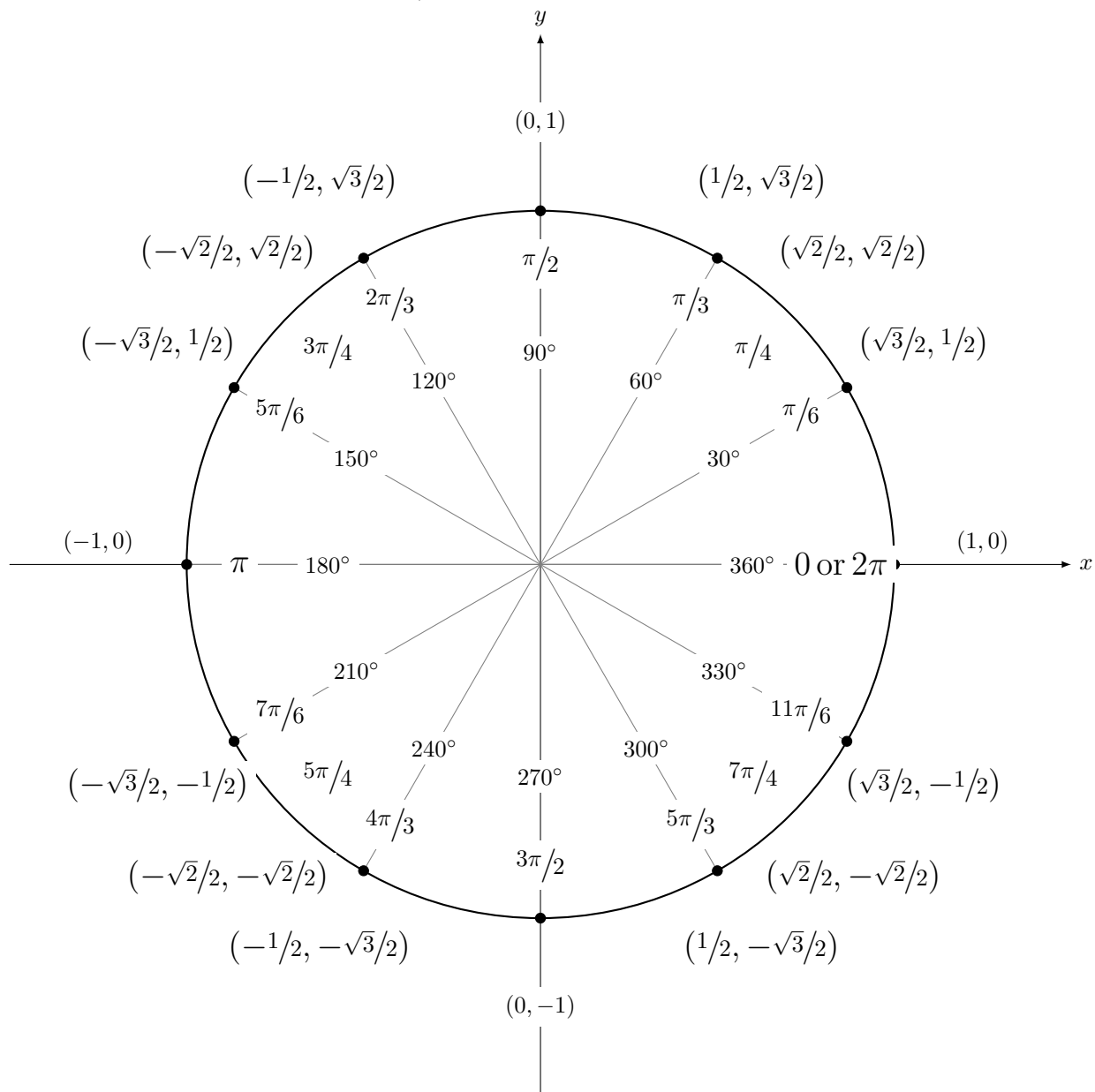
## Inverse Functions

$$\begin{aligned}
 \cos^{-1}(z) &= -i \log(z + i(1-z^2)^{1/2}) \\
 \sin^{-1}(z) &= -i \log(iz + (1-z^2)^{1/2}) \\
 \tan^{-1}(z) &= \frac{1}{2i} \log \frac{i-z}{i+z} \\
 \cosh^{-1}(z) &= \log(z + (z^2-1)^{1/2}) \\
 \sinh^{-1}(z) &= \log(z + (1+z^2)^{1/2}) \\
 \tanh^{-1}(z) &= \frac{1}{2} \log \frac{1+z}{1-z}
 \end{aligned}$$

## Binomial Theorem

$$(1+x)^\alpha = \sum_{n=0}^{\infty} \binom{\alpha}{n} x^n \text{ for } |x| < 1, \alpha \in \mathbb{C}$$

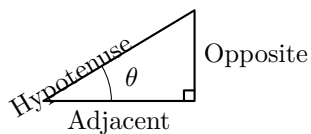
Unit circle (taken from <http://www.texample.net/tikz/examples/unit-circle/> by Supreme Aryal, under creative commons license 2.5)



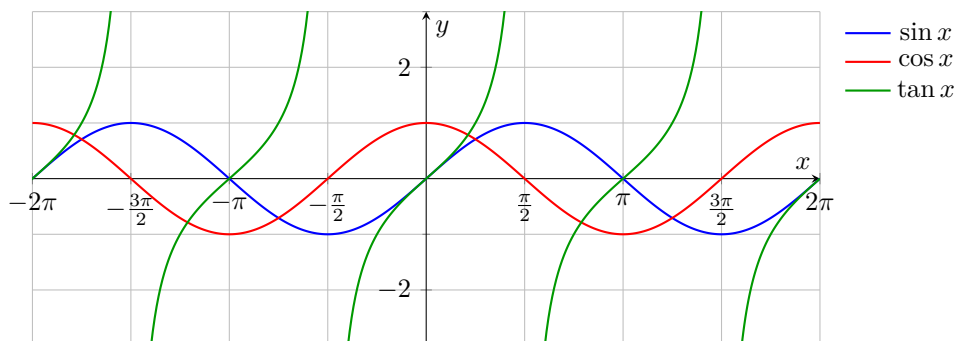
$$\sin \theta = \frac{\text{Opposite}}{\text{Hypotenuse}}$$

$$\cos \theta = \frac{\text{Adjacent}}{\text{Hypotenuse}}$$

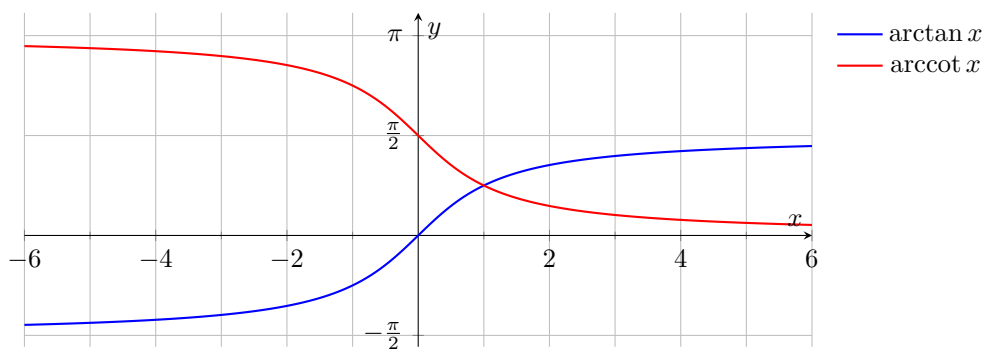
$$\tan \theta = \frac{\text{Opposite}}{\text{Adjacent}}$$



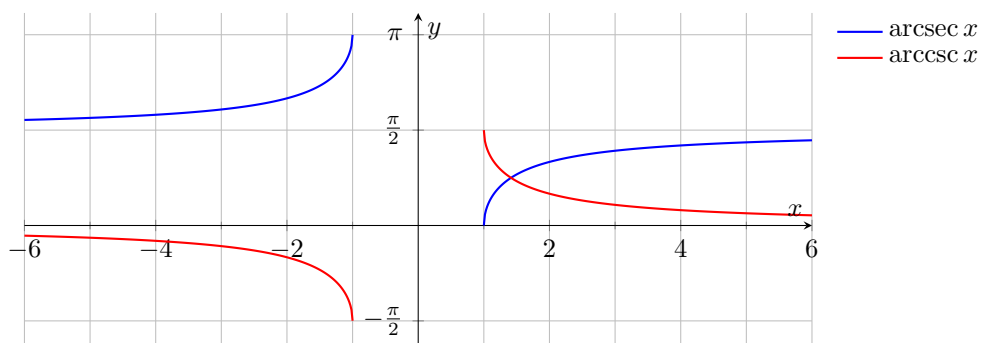
Trigonometric functions



Inverse tangent and cotangent



Inverse secant and cosecant



Inverse sine and cosine

